



Build with an AI-native DB:

An overview of Weaviate



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What is **Weaviate** ?

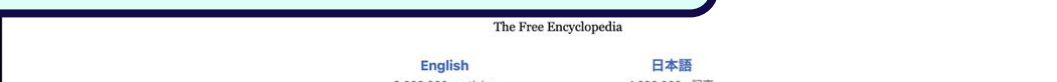
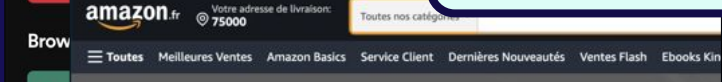
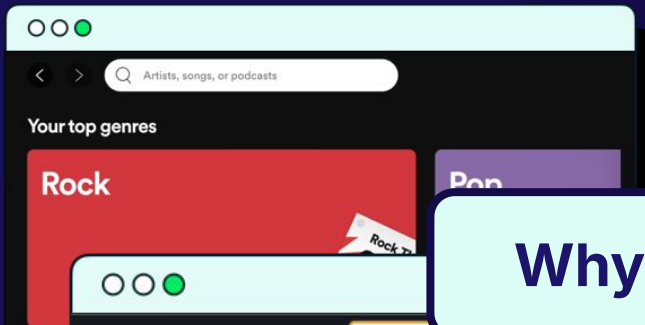
An introduction to vectors and vector DBs



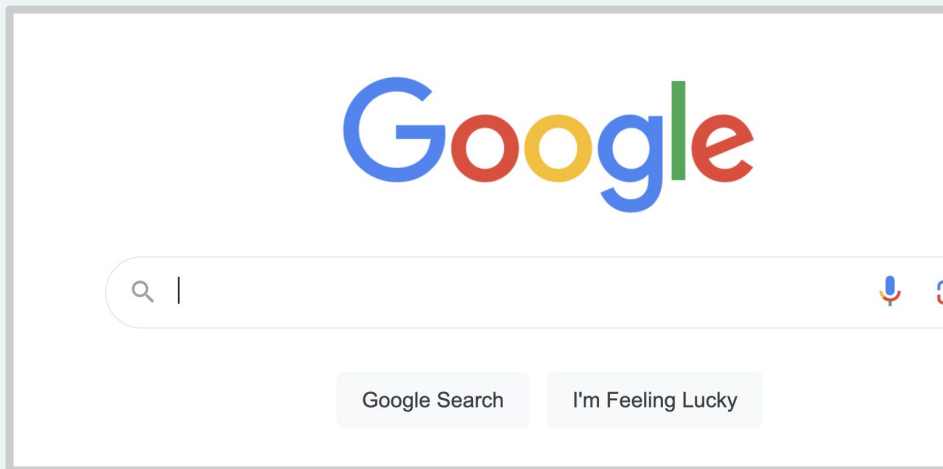
Everything we do
starts with search



Why is search so ubiquitous?



We are spoiled





The problem of *effective search*

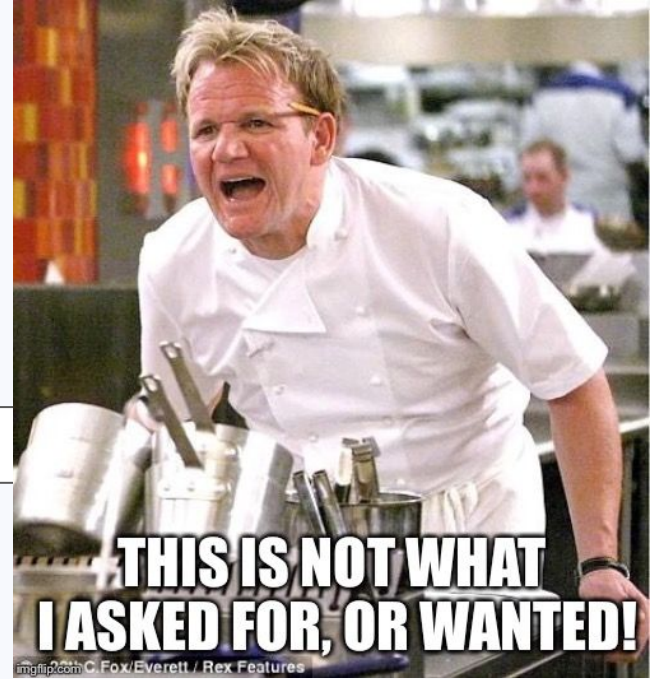
Traditional search has shortcomings

How do airplanes fly?



[Why you should fly with ExpensiveAir's airplanes!](#)

...fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. **How to book: ...**



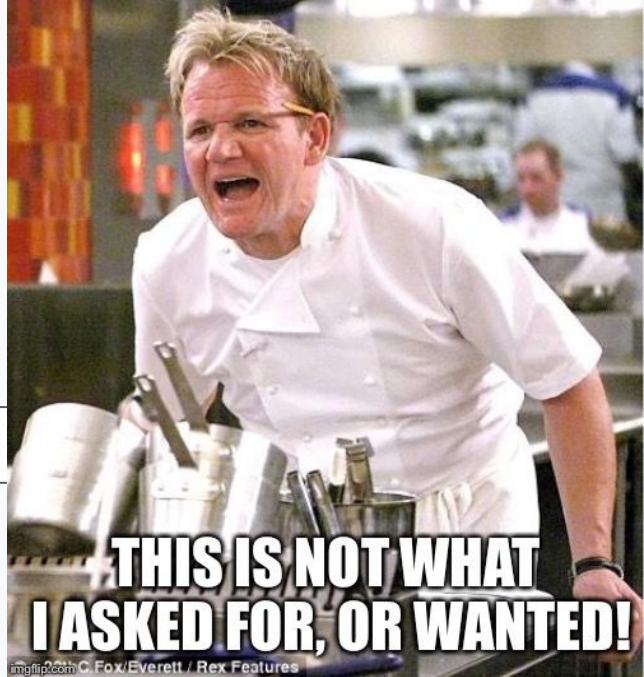
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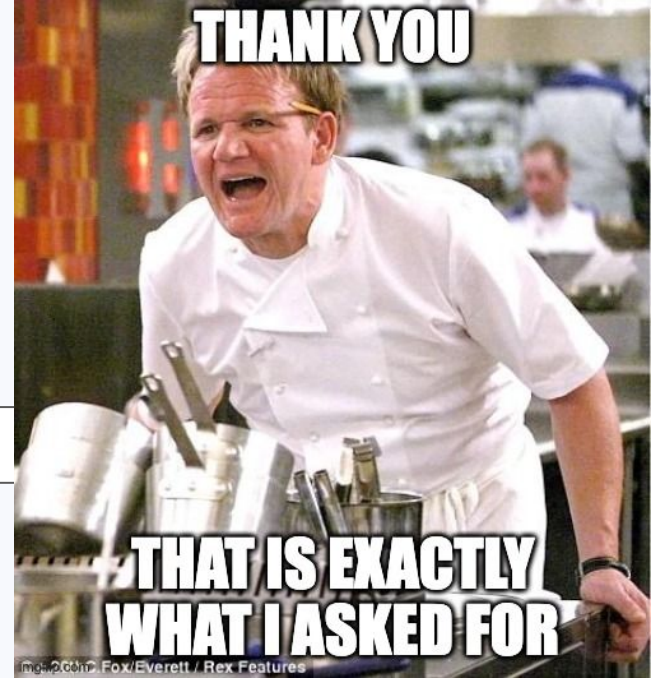
Semantic search to the rescue!

How do airplanes fly?



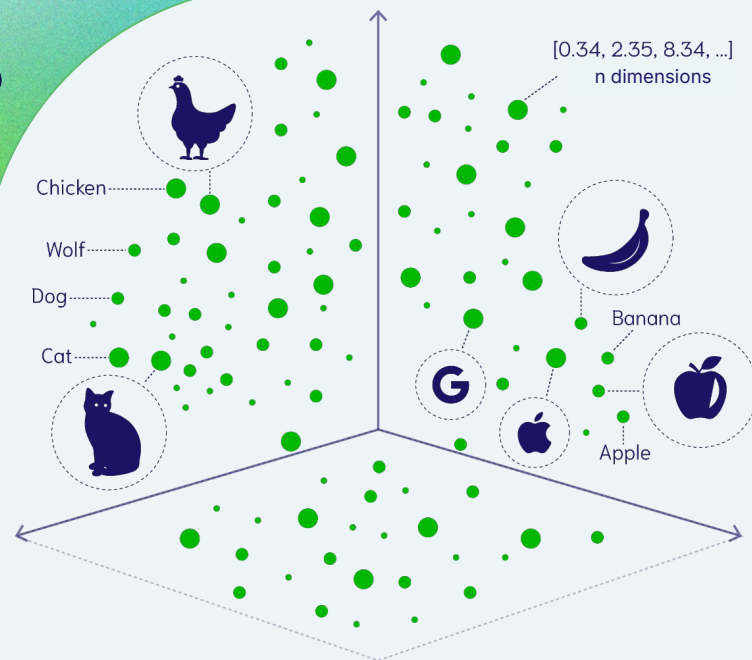
[Dynamics of Flight - NASA](#)

Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases.



How does this work?

Vectors!





What is a vector?

A vector is a set of numbers

That represents some
meaning

Like

`[1, 0]`

or

`[0.513, 0.155, 0.983, 0.001,
0.932]`

or

`[0.0009420722, 0.020158706,
-0.03939992, 0.018441677, ...,
-0.025270471, -0.056622636, ...]`

Numbers represent meaning?

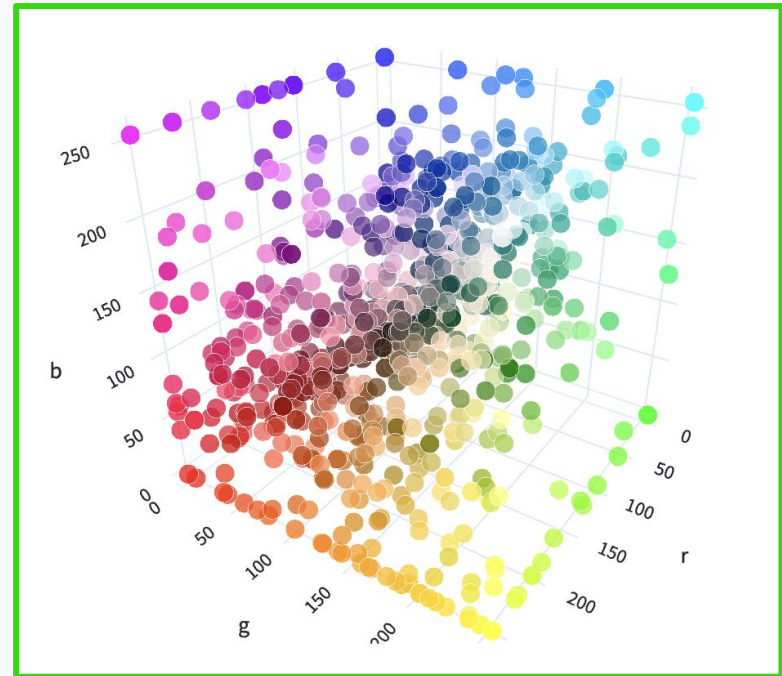
Here's an example:

RGB numbers represent colors, like:

$(255, 0, 0)$ = red

$(80, 200, 120)$ = emerald

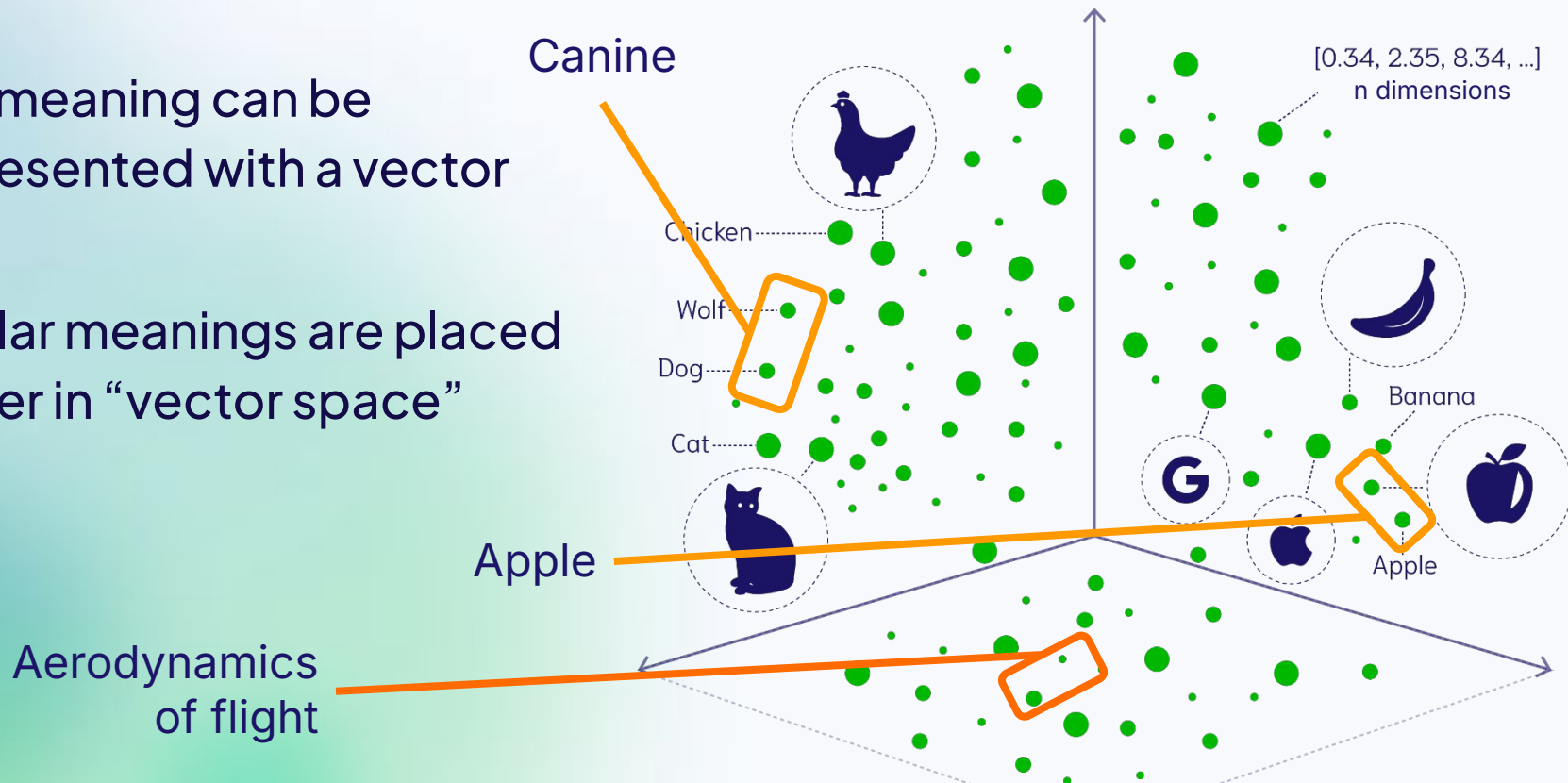
Each number is a dial
for (red, green, blue) ness.



Numbers represent meaning

Any meaning can be represented with a vector

Similar meanings are placed closer in “vector space”





Where do vectors come from?

Machine Learning models



[0.23, 0.45, ..., 0.84, 0.23]

Vector Embeddings



“Fly with ExpensiveAir’s modern fleet of airplanes to over 4000 destinations. How to book...”



[0.23, 0.45, ..., 0.84, 0.23]



Vector Embeddings

“Fly with ExpensiveAir’s modern fleet of airplanes to over 4000 destinations. How to book...”

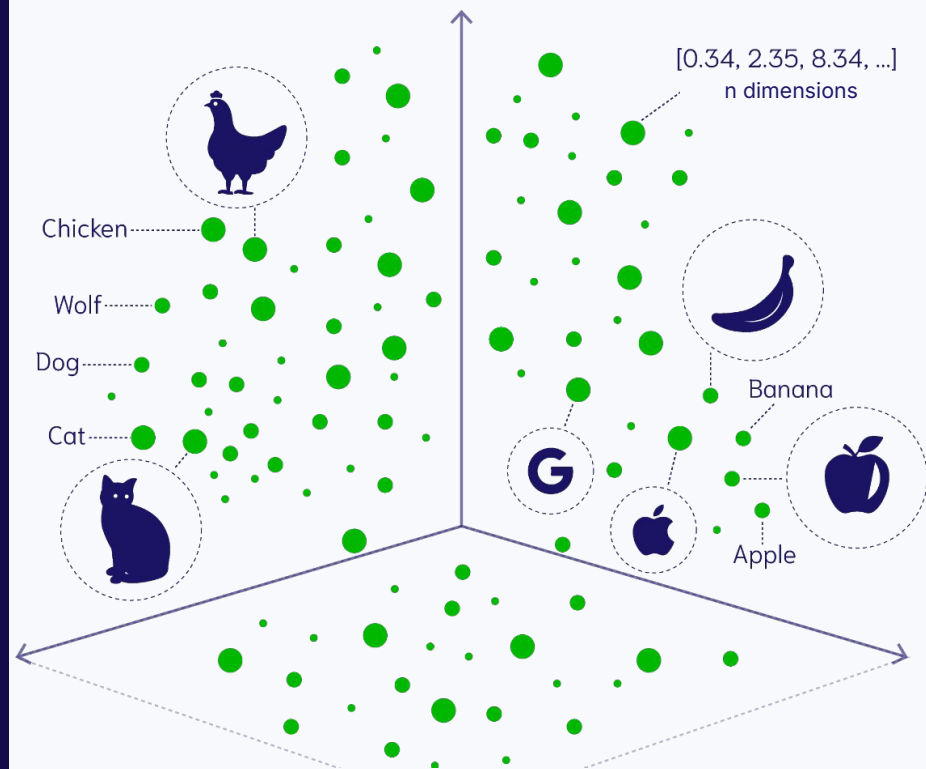
Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases.



[0.23, 0.45, ..., 0.84, 0.23]

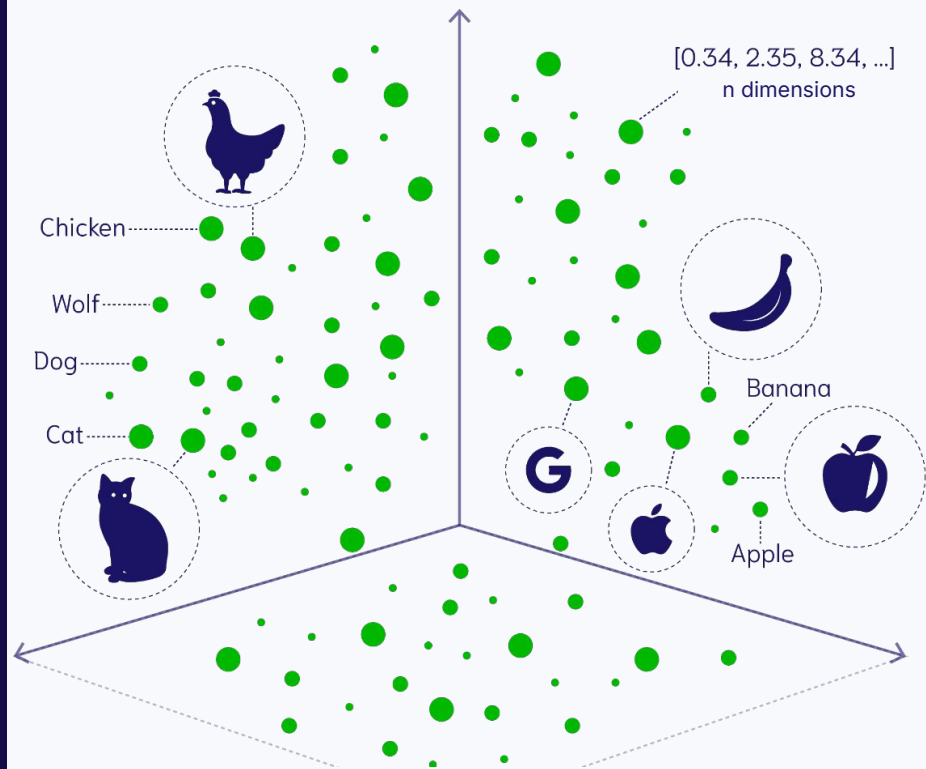
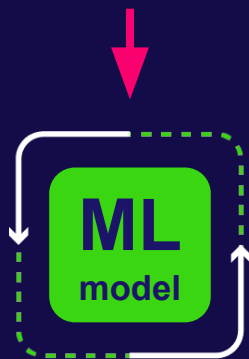
[0.26, 0.31, ..., 0.12, 0.44]

Vector Space



Vector Search

How do airplanes fly?

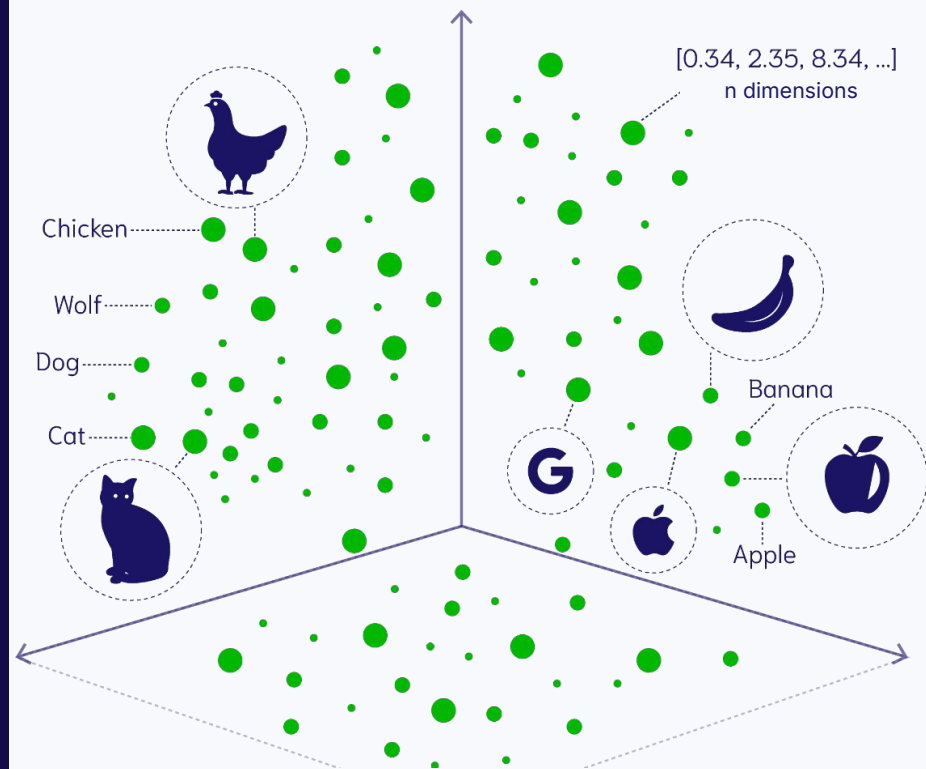


Vector Search

How do airplanes fly?



[0.24, 0.36, ..., 0.16, 0.46]

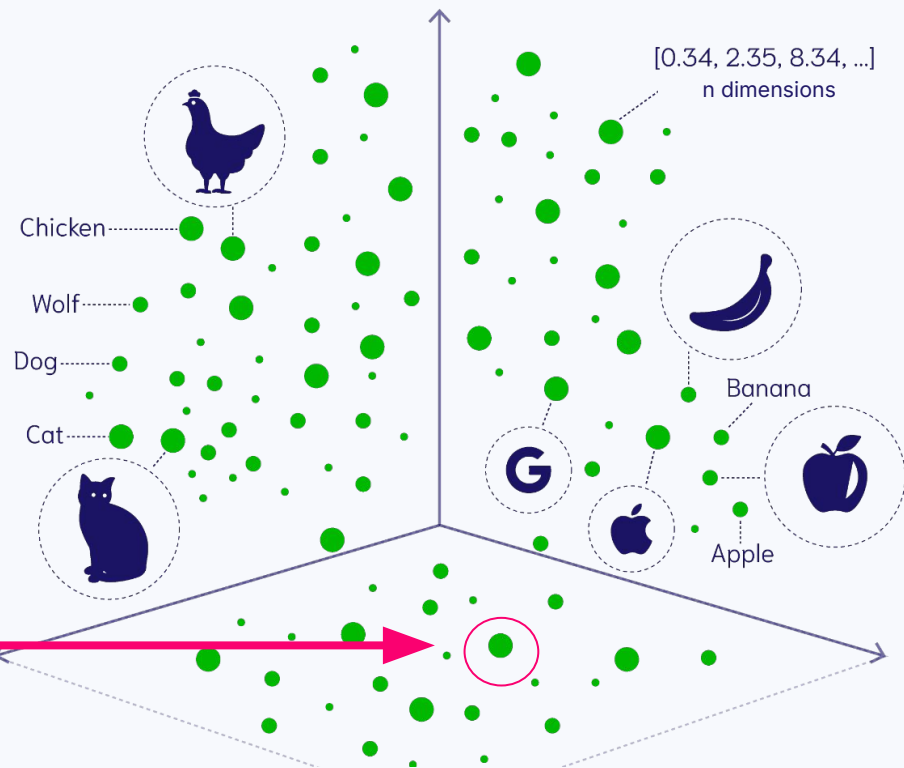


Vector Search

How do airplanes fly?



[0.24, 0.36, ..., 0.16, 0.46]



Vector Search

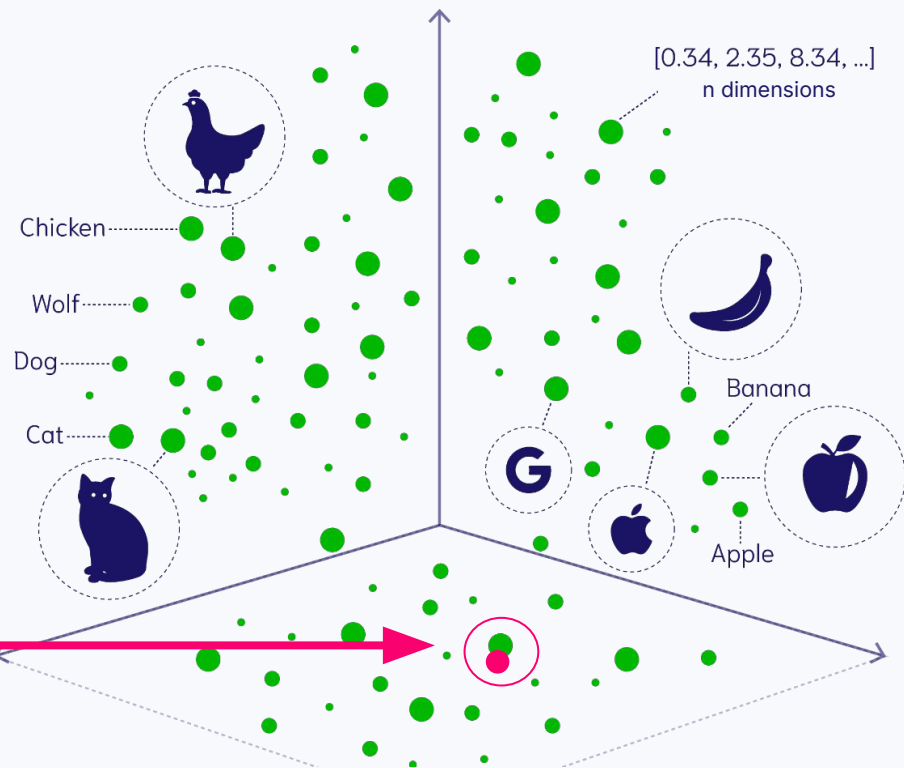
How do airplanes fly?



[0.24, 0.36, ..., 0.16, 0.46]

[0.26, 0.31, ..., 0.12, 0.44]

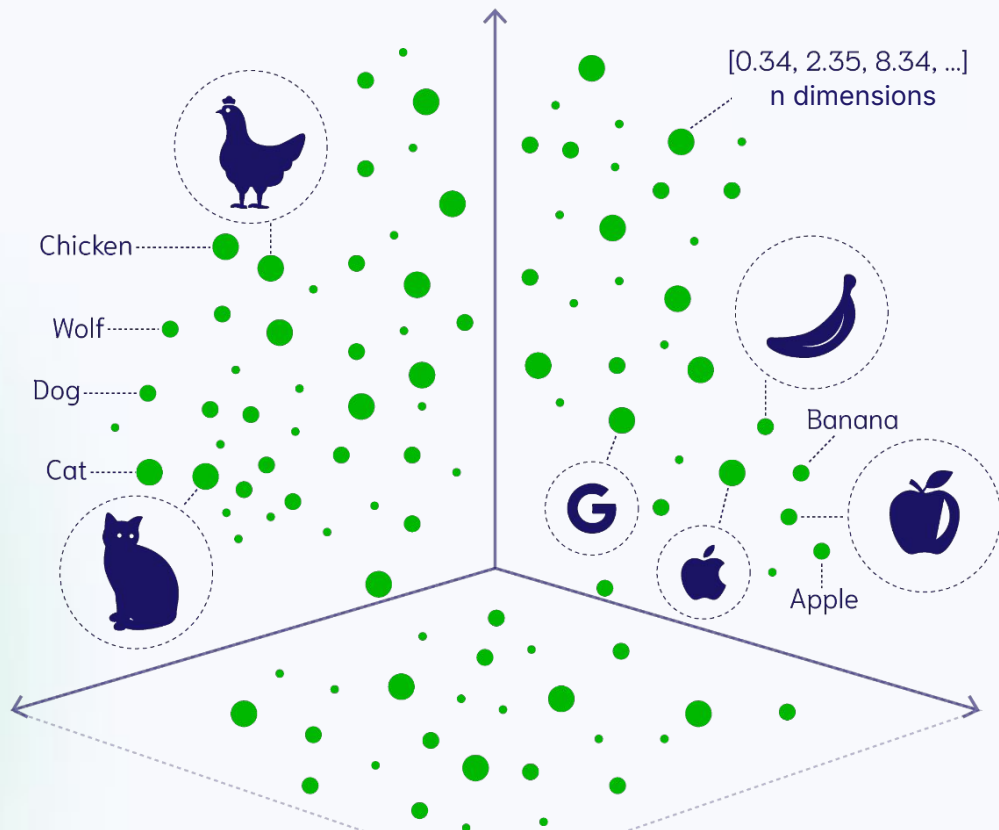
The NASA article



What is a vector database?

A database whose primary **data organization** is by meaning (vectors)

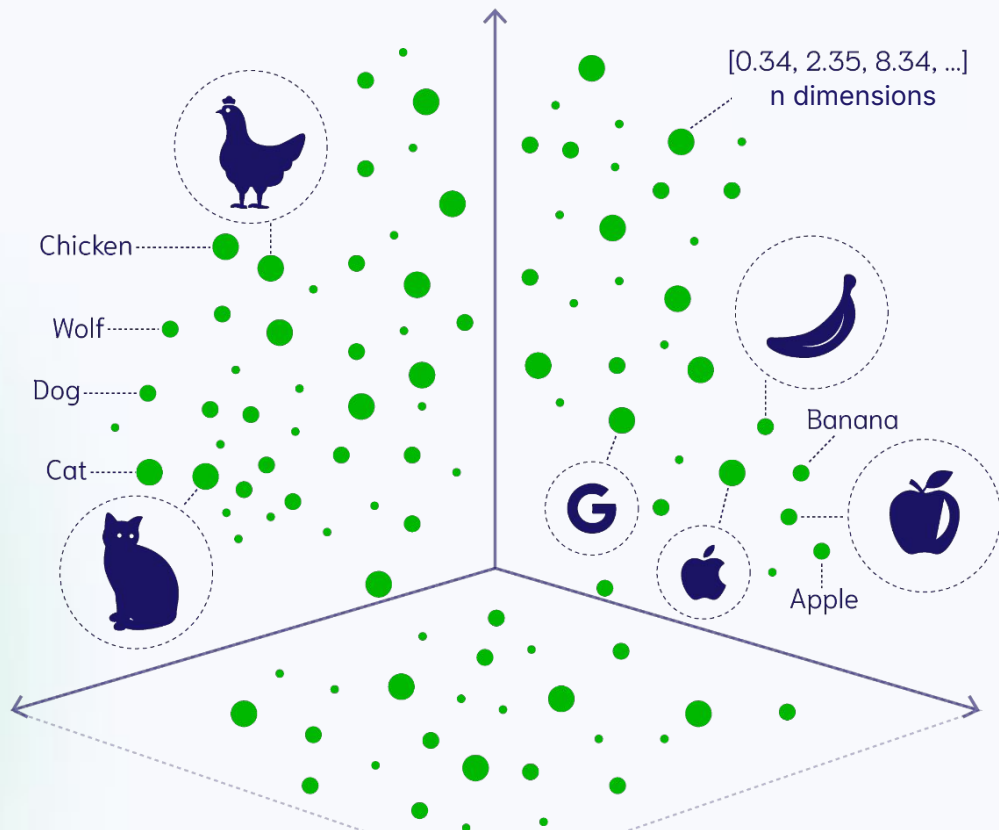
And optimized for data management and queries for using vectors **at scale**.
(billions)



What is an AI-native database?

Vectors and AI-native workflow as first-class citizens

- AI model integrations
- Hybrid search
- Scalability (multi-tenancy, quantization)
- Developer experience





Better search

Use semantic, keyword and hybrid searches
to get the right data, faster



Keyword, Vector or Hybrid?

Differences between search types

Search: example

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

Search: example

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

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Explains aerodynamics

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

how are they compared / searched?

Keyword search (BM25)

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

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"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

Keyword search:

match keywords and calculate a score based on counts & typical frequency (TF-IDF)

Keyword search (BM25)

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

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Advantages of flight

Query:

"How do airplanes fly?"

score: 4.8
(for example)

Keyword search:

match keywords and calculate a score based on counts & typical frequency (TF-IDF)



Keyword search results

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

Rank: 1

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

Not in results

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Rank: 2

Query:

"How do airplanes fly?"

finds documents with the most important keyword matches

Keyword search:

match keywords and calculate a score based on counts & typical frequency (TF-IDF)

When to use...

Keyword search

When exact matches important.

Examples:

- Domain-specific data (jargon)
- Product / part numbers
- URL matches

Keyword search syntax



```
1 products = client.collections.get("Product")
2
3 response = products.query.bm25(
4     "How do airplanes fly?",
5     limit=2
6 )
```

Vector search

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

Vector search:

Convert each object into a vector and compare their numeric similarity

Vector search

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

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"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

[0.26, 0.31, ..., 0.12, 0.44]

[0.23, 0.45, ..., 0.84, 0.23]

similarity: 0.4
(for example)

Vector search:

Convert each object into a vector and compare their numeric similarity



Vector search results

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

Rank: 3

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

Rank: 1

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Rank: 2

Query:

"How do airplanes fly?"

finds documents with the highest similarity in meaning

Vector search:

Convert each object into a vector and compare their numeric similarity

When to use...

Vector search

When robustness is important, or cross-lingual or cross-modal search is desired.

Examples:

- Descriptions
- Image to text / image to image
- Product reviews

Vector search syntax



```
1 products = client.collections.get("Product")
2
3 response = products.query.near_text(
4     "How do airplanes fly?",
5     limit=2
6 )
```



Hybrid search

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Query:

"How do airplanes fly?"

Hybrid search:

Perform both a vector search and keyword search & combine the results for a final ranking

Hybrid search

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

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Advantages of flight

Query:

"How do airplanes fly?"

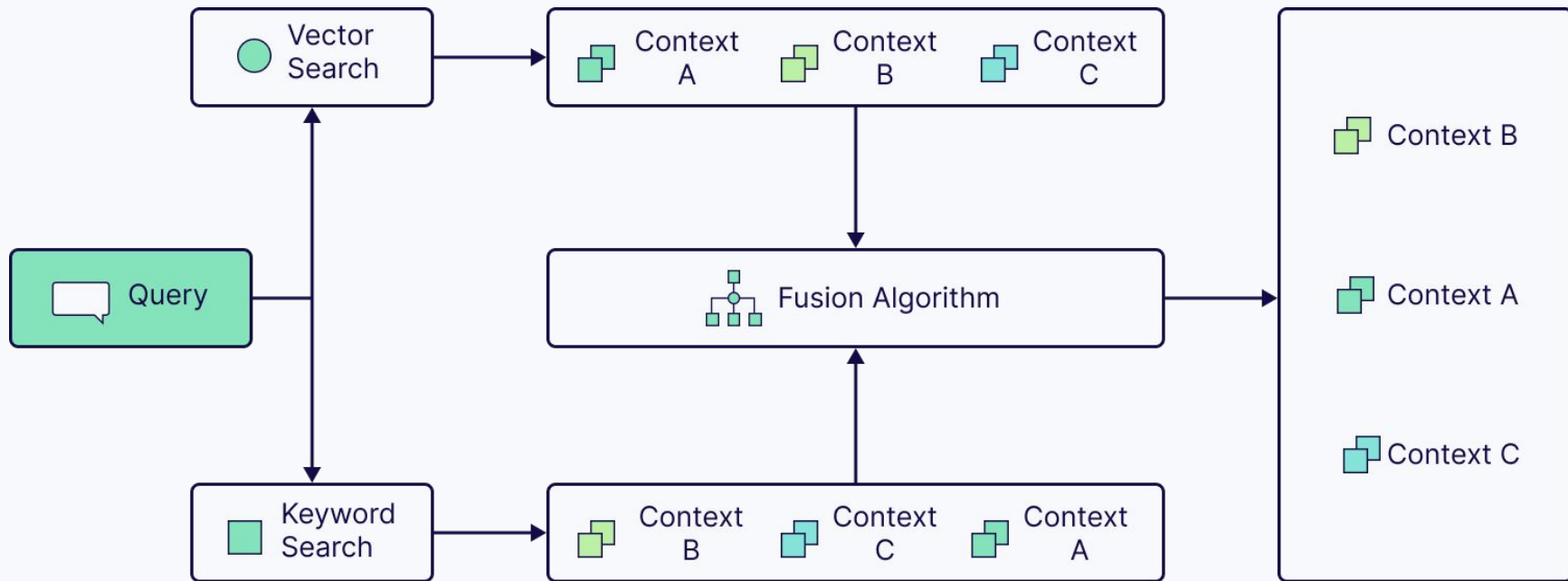
vector search similarity: 0.4
keyword search score: 4.8

Calculate overall score & rank against others

Hybrid search:

Perform both a vector search and keyword search & combine the results for a final ranking

Hybrid Search





Hybrid search results

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Rank: 3

Query:

"How do airplanes fly?"

mediocre results in both
are not rewarded

Hybrid search:

Perform both a vector search and keyword search & combine the results for a final ranking



Hybrid search results

Source documents

"Come and fly with ExpensiveAir's modern fleet of airplanes to over 4000 destinations. How to book a flight with ExpensiveAir?..."

Ad for an airline

Rank: 1/2

"Airplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases."

Explains aerodynamics

Rank: 1/2

"Flying is a great way to travel long distances quickly. Airplanes are able to carry large numbers of passengers and cargo over long distances."

Advantages of flight

Rank: 3

Query:

"How do airplanes fly?"

hybrid results reward high rankings in at least one of keyword/vector

mediocre results in both are not rewarded

Hybrid search:

Perform both a vector search and keyword search & combine the results for a final ranking

When to use...

Hybrid search

Best-of-both-worlds search. Surfaces top results from both keyword & vector searches

Good results reported “in the wild” from users due to resilience.

Hybrid search syntax



```
1 products = client.collections.get("Product")
2
3 response = products.query.hybrid(
4     "How do airplanes fly?",
5     limit=2,
6     alpha=0.5
7 )
```



AI-native workflows

About Weaviate's AI-native developer experience,
integrating embedding and genAI models directly



API-based

Model provider	Embeddings	Generative AI	Others
Anthropic	-	Text	-
Anyscale	-	Text	-
AWS	Text	Text	
Cohere	Text, Multimodal	Text	Reranker
Databricks	Text	Text	-
FriendliAI	-	Text	-
Google	Text, Multimodal	Text	-
Hugging Face	Text	-	-
Jina AI	Text, Multimodal	-	Reranker
Mistral	Text	Text	-
NVIDIA	Text, Multimodal	Text	Reranker
OctoAI (Deprecated)	Text	Text	-
OpenAI	Text	Text	-
Azure OpenAI	Text	Text	-
Voyage AI	Text, Multimodal	-	Reranker
Weaviate	Text	-	-
xAI	-	Text	-

Out-of-the-box integrations

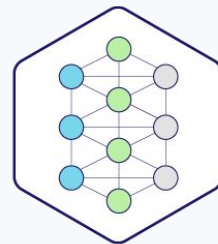
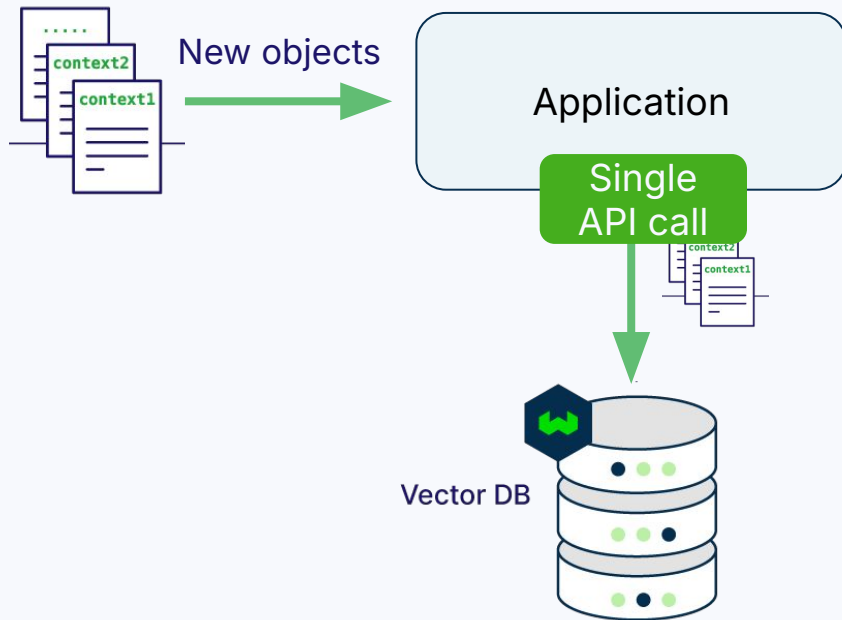
Locally hosted

Model provider	Embeddings	Generative AI	Others
GPT4All	Text	-	-
Hugging Face	Text, Multimodal (CLIP)	-	Reranker
Meta ImageBind	Multimodal	-	-
Ollama	Text	Text	-



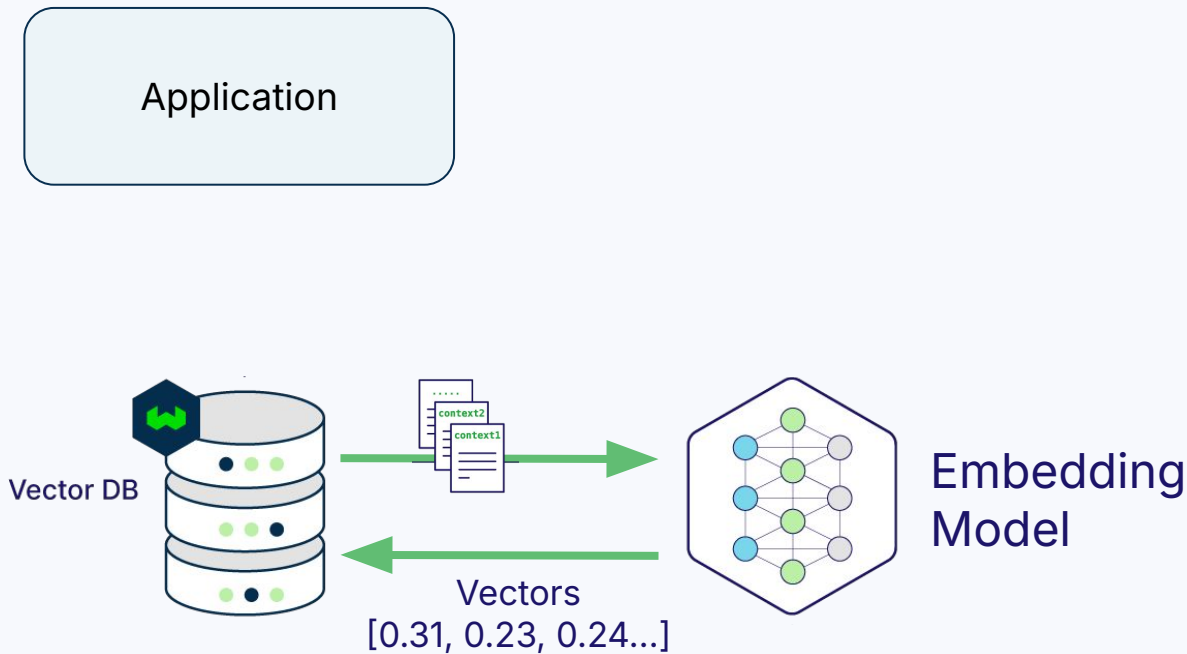
Embedding model integration

Data operations (i.e. data import)

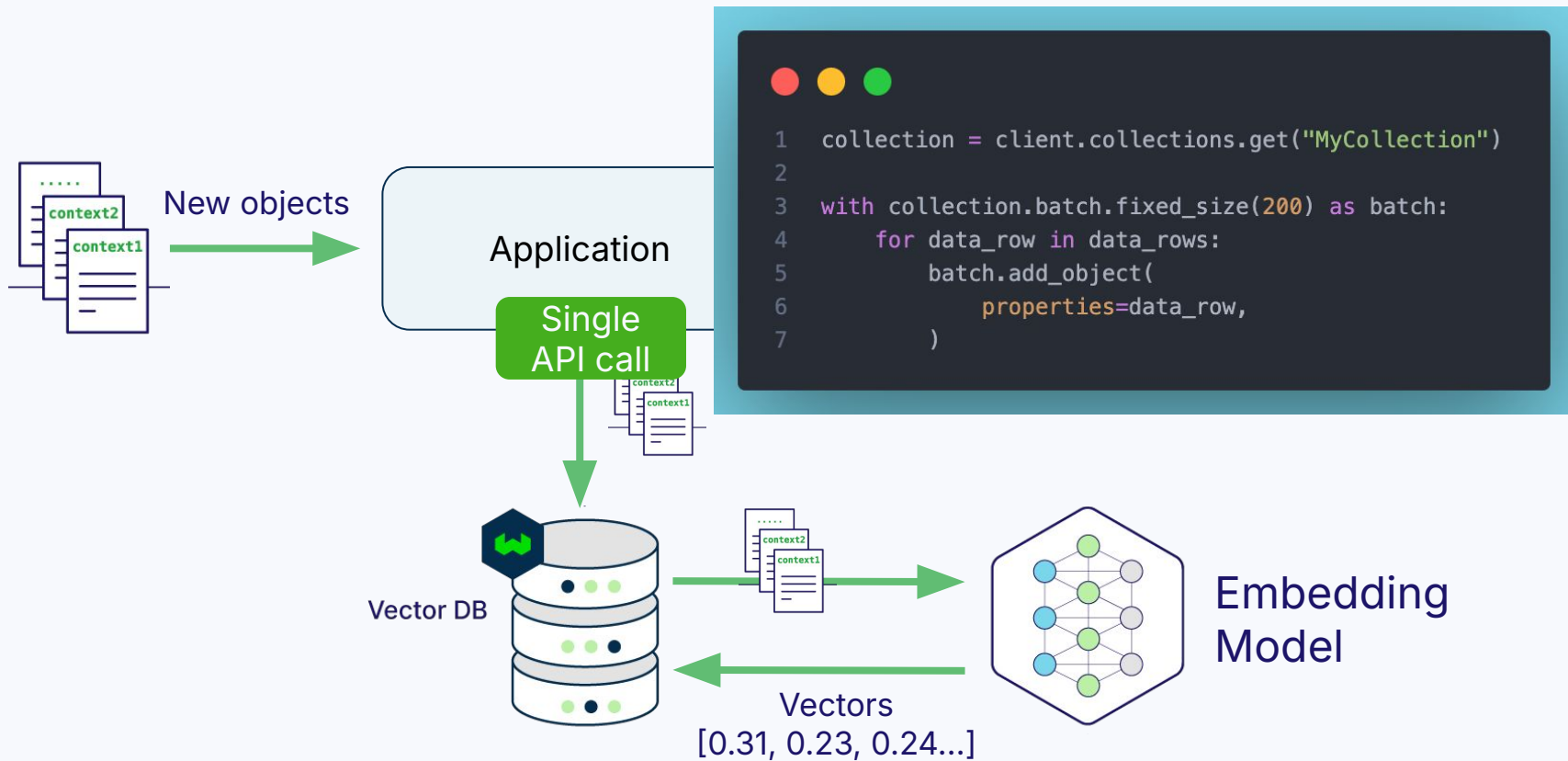


Embedding
Model

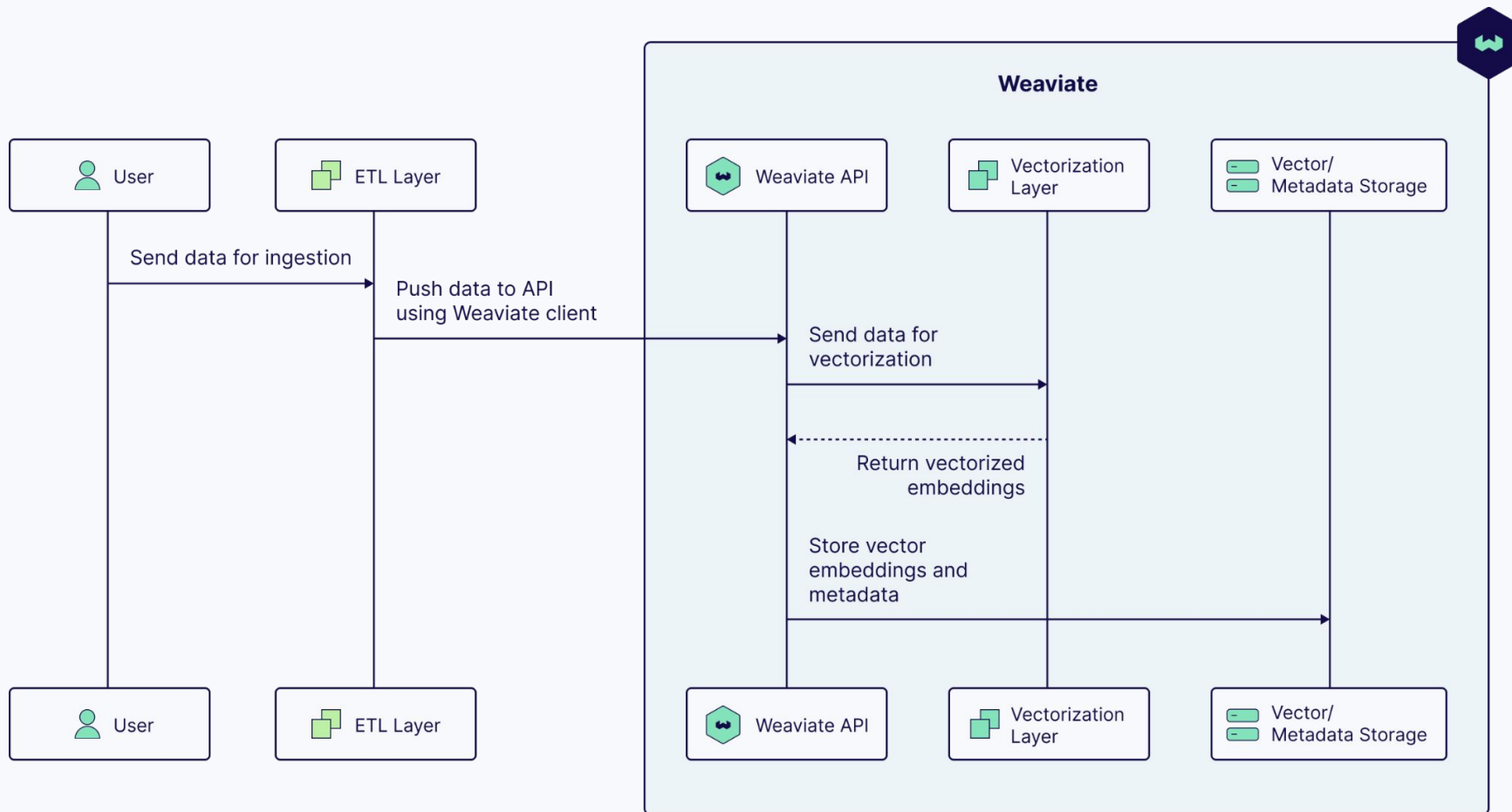
Data operations (i.e. data import)



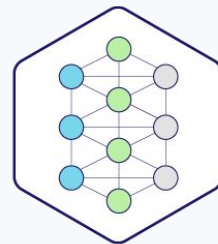
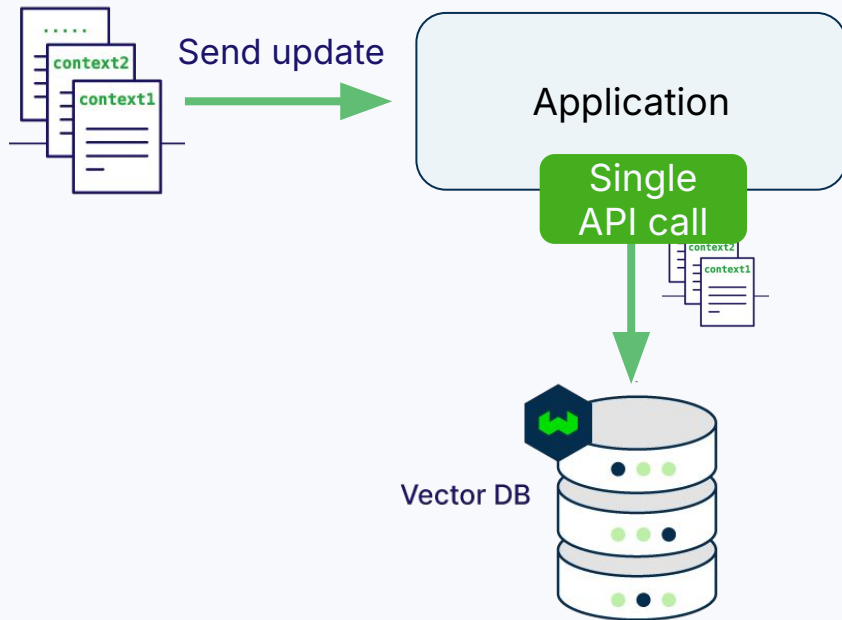
Data operations (i.e. data import)



Data Ingestion

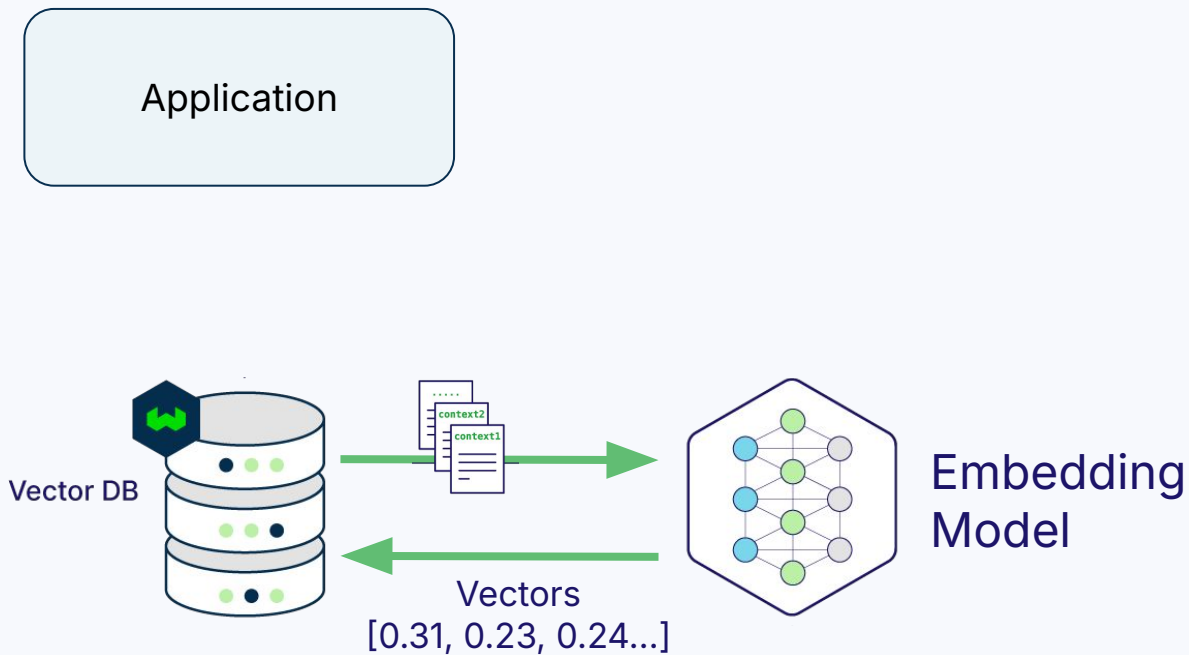


Data operations (i.e. update)

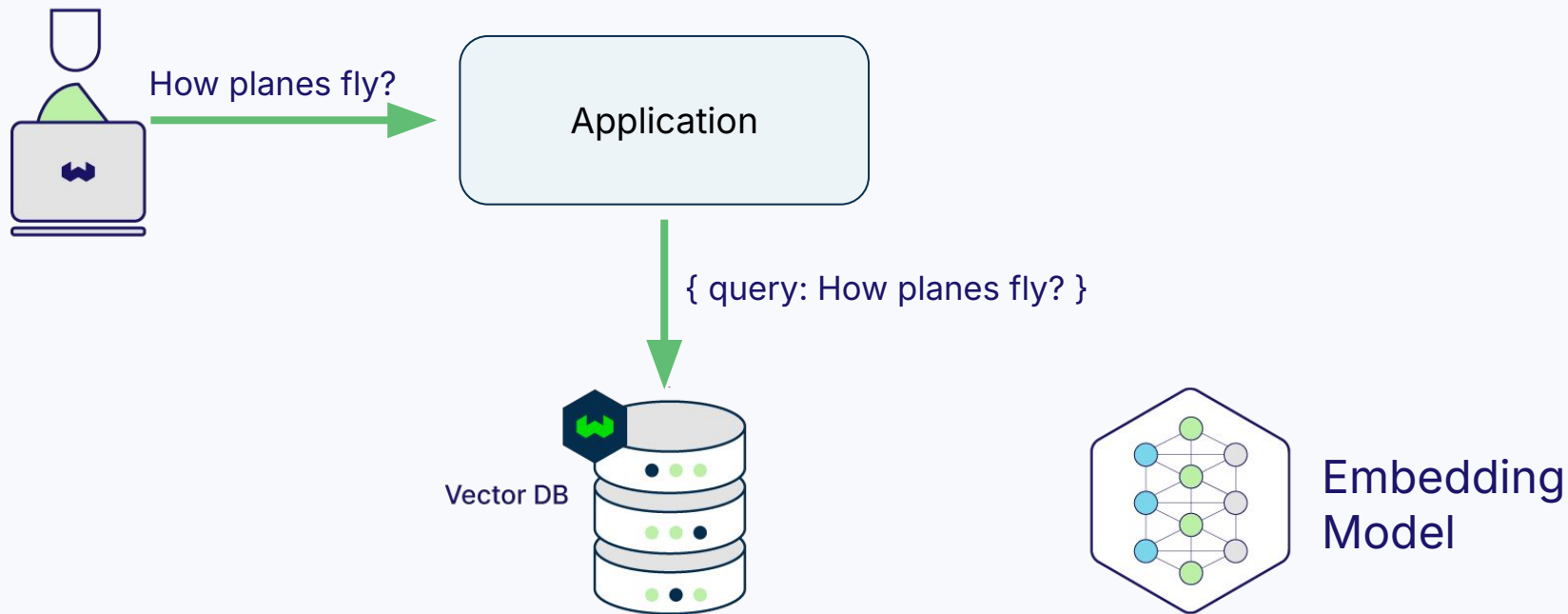


Embedding Model

Data operations (i.e. update)



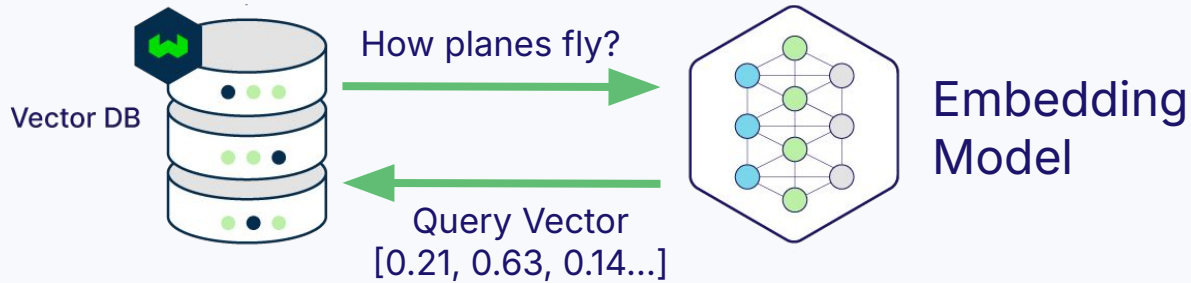
Query



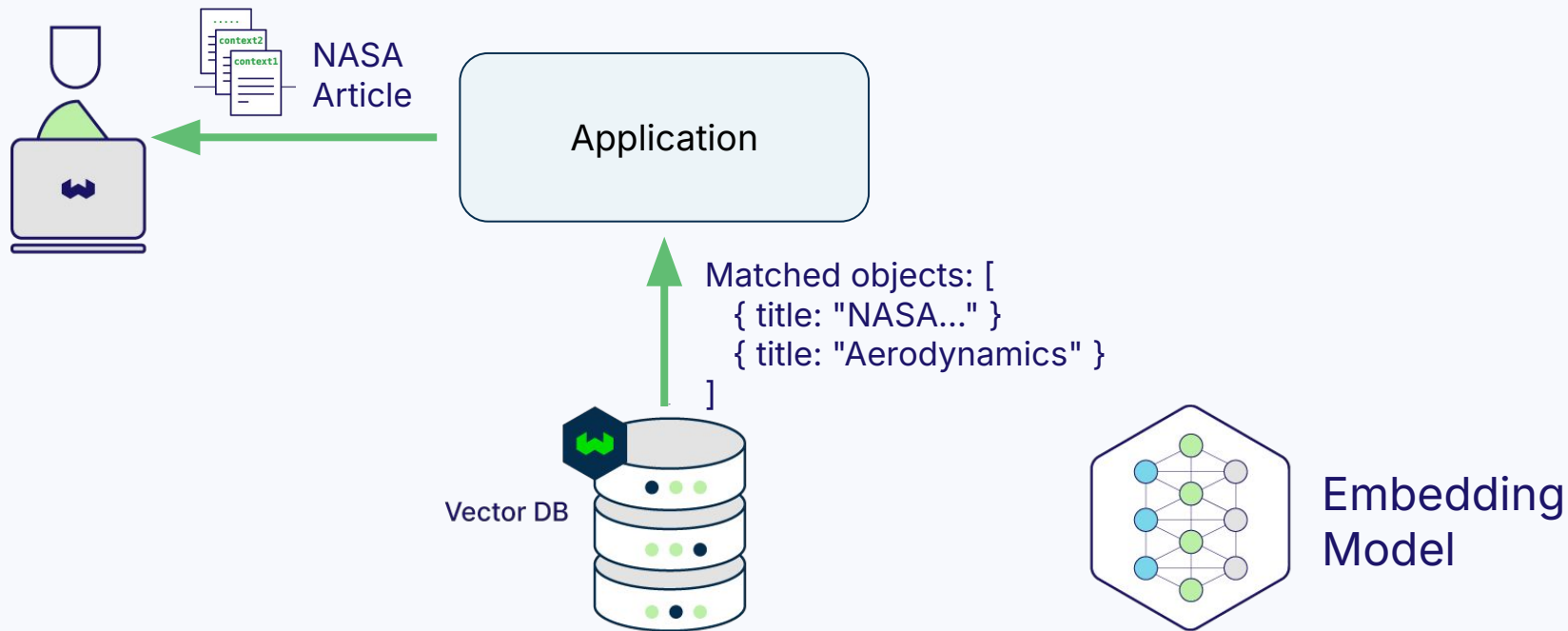
Query



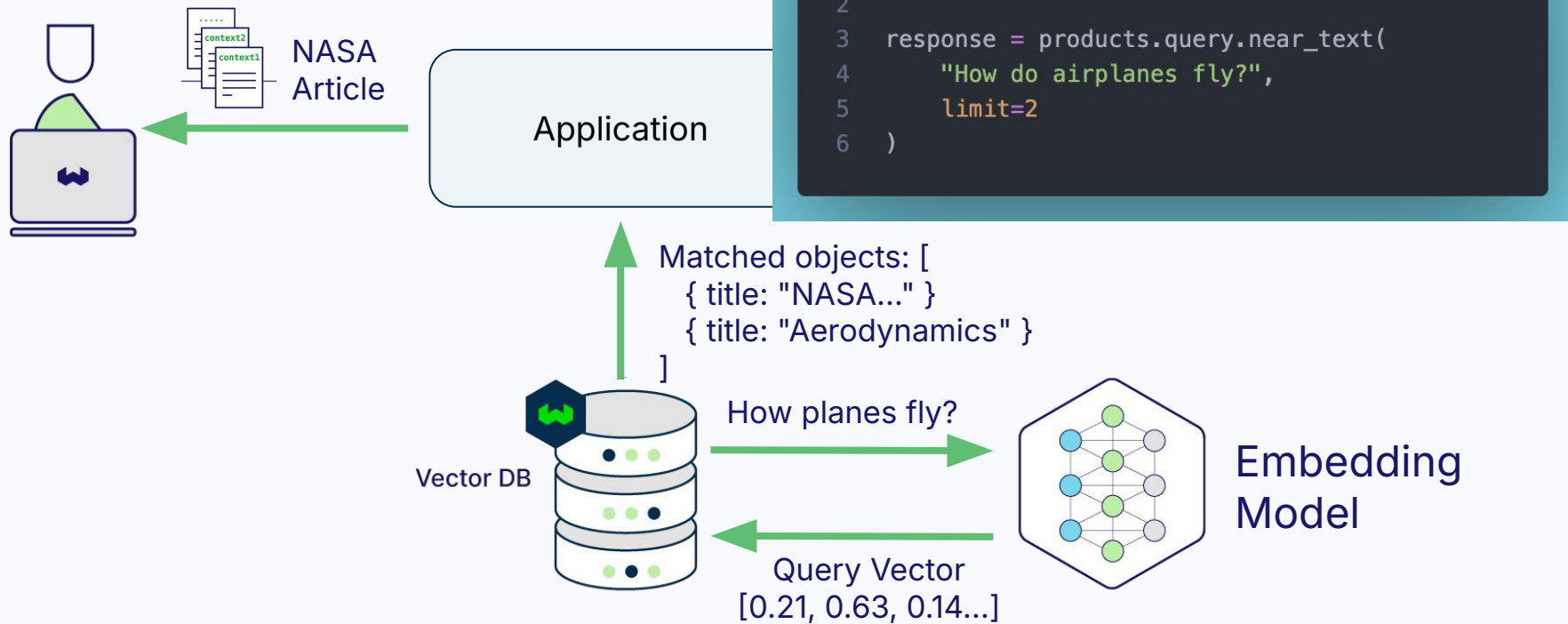
Application



Query



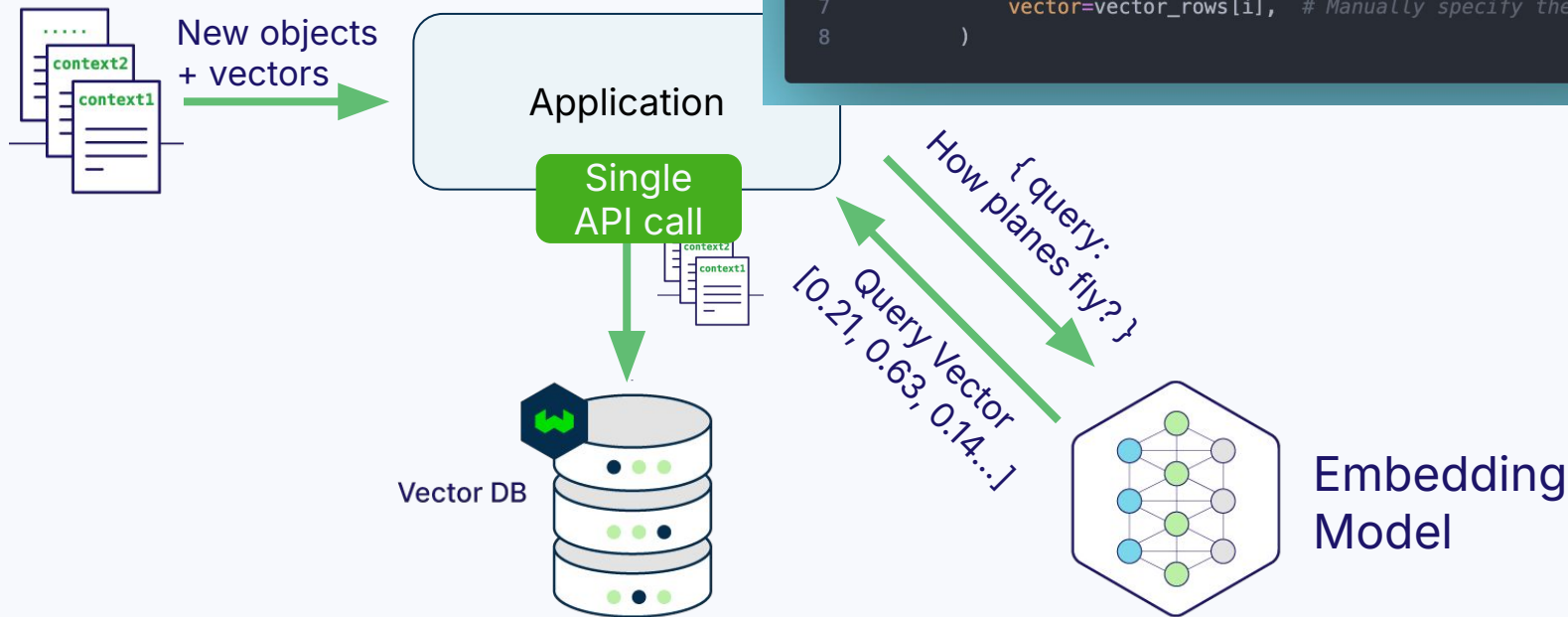
Query



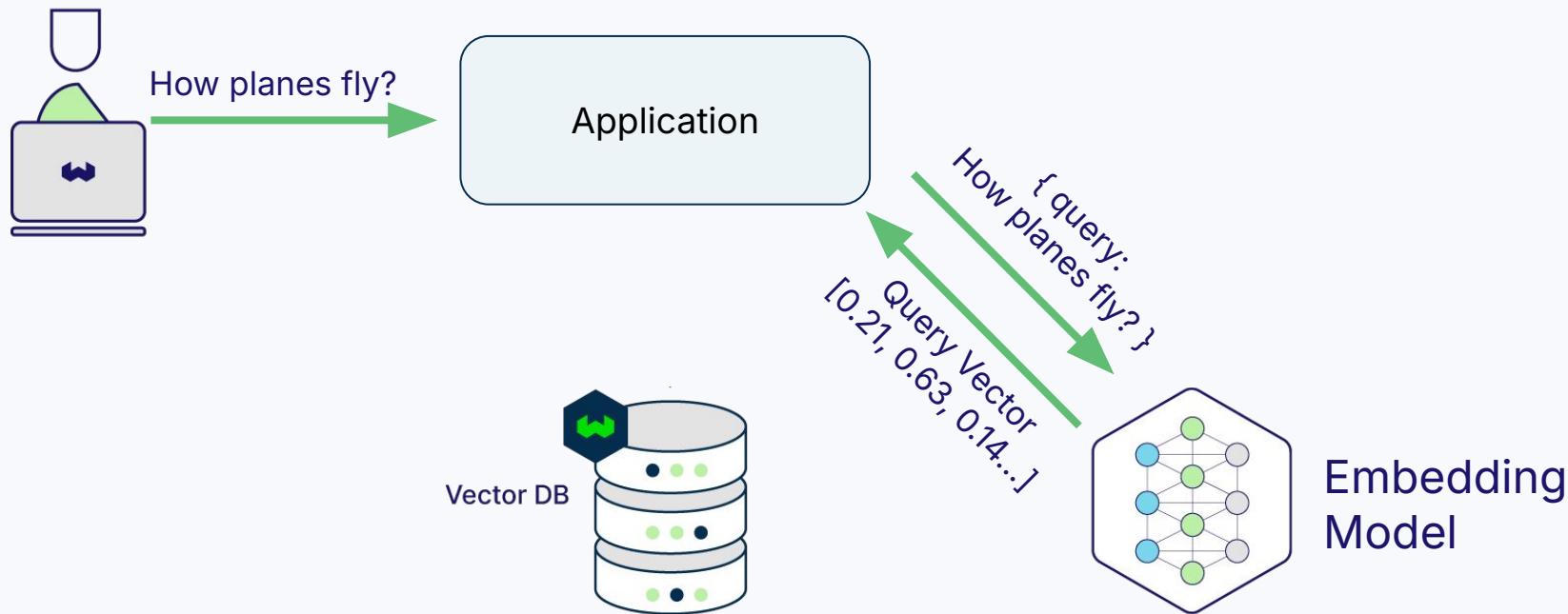


Bring your own vectors

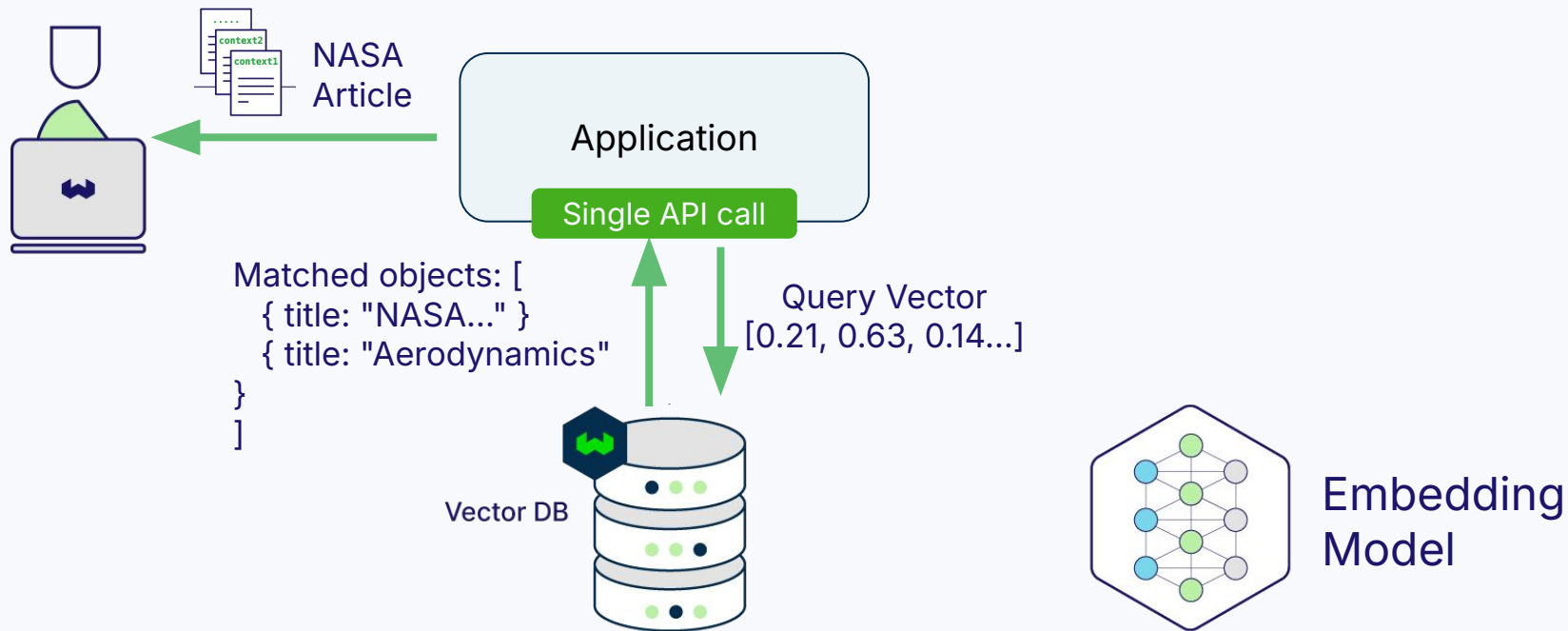
Import Data with Vectors (no vectorizer)



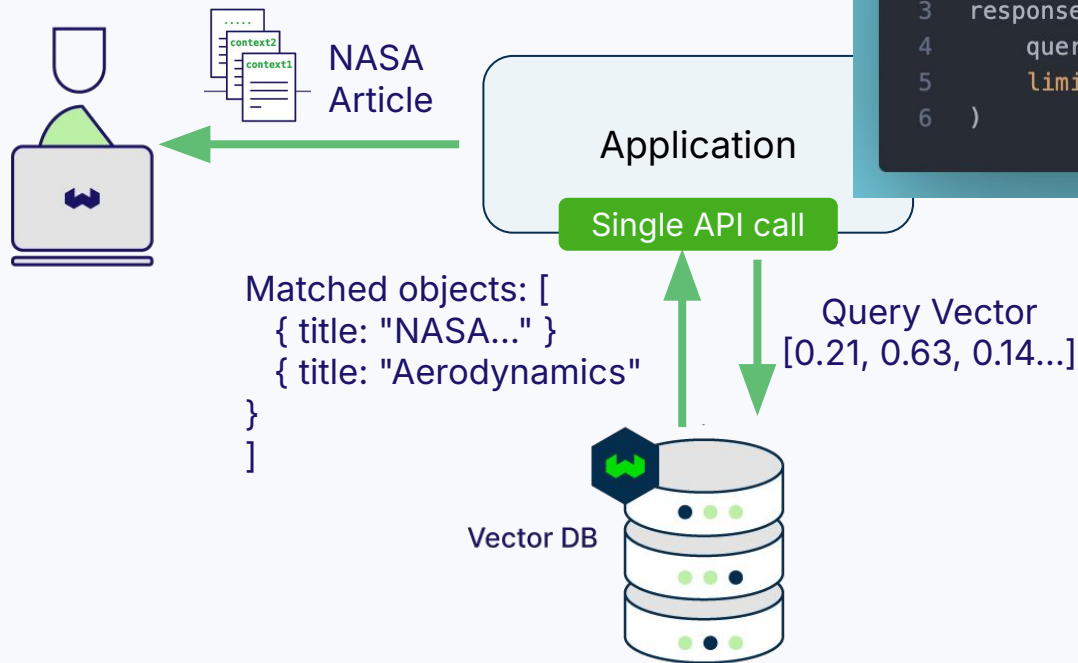
Query (no vectorizer)



Query (no vectorizer)

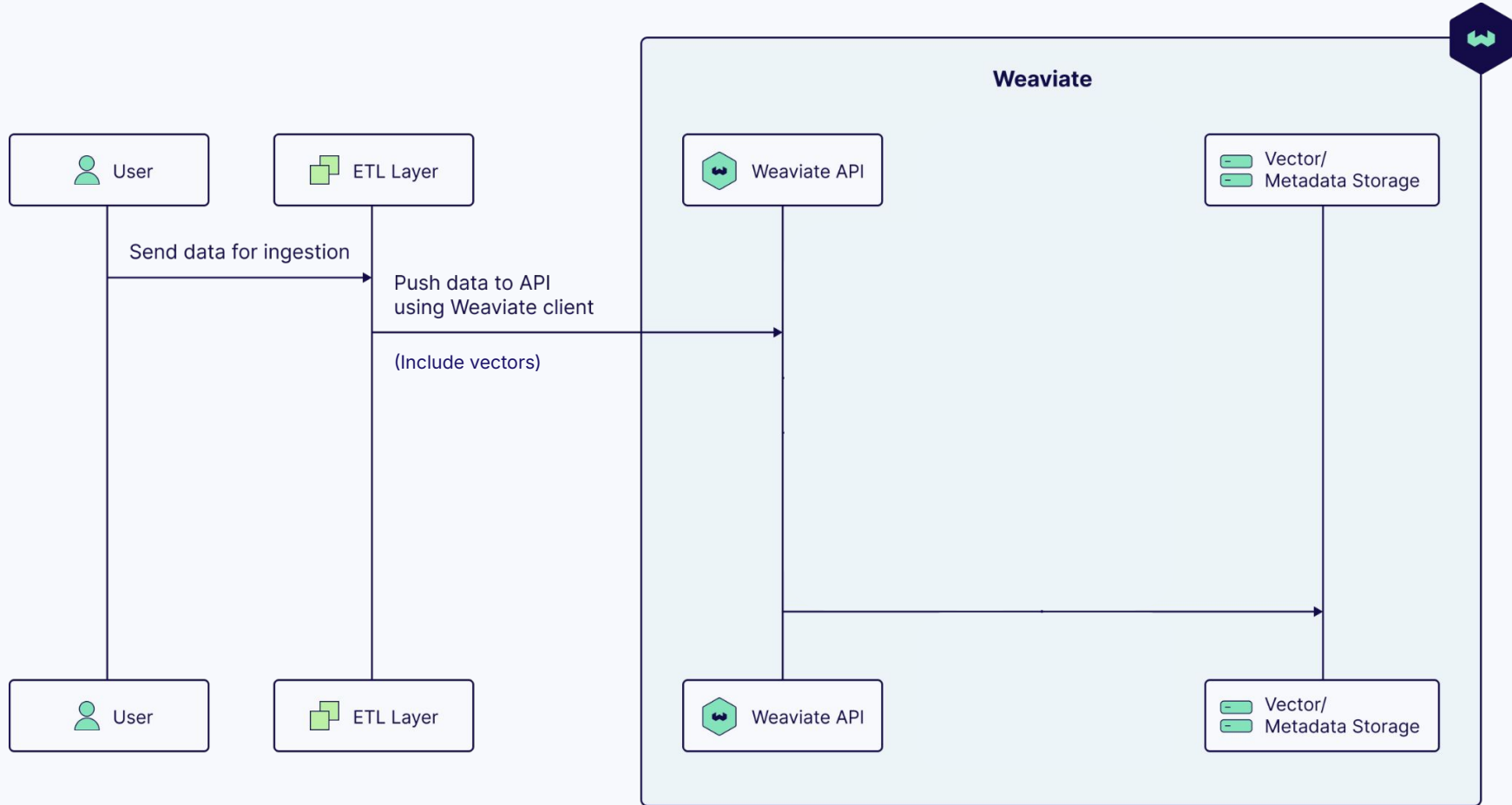


Query (no vectorizer)



```
1 products = client.collections.get("Product")
2
3 response = products.query.near_vector(
4     query_vector, # Manually obtained query vector
5     limit=2
6 )
```

Data Ingestion





GenAI model integration

Vector Database

Large Language Model

One Key Difference





**Vector Database
(Stateful)**

Large Language Model
(Stateless)



Vector Database (Stateful)

- Insert data into the database, and generate vectors.

Large Language Model (Stateless)

- The knowledge base is trained into the model.



Vector Database (Stateful)

- Insert data into the database, and generate vectors.
- Data changes on the fly.

Large Language Model (Stateless)

- The knowledge base is trained into the model.
- The model doesn't learn on the fly.



Vector Database (Stateful)

- Insert data into the database, and generate vectors.
- Data changes on the fly.
- CRUD operations to update the state.

Large Language Model (Stateless)

- The knowledge base is trained into the model.
- The model doesn't learn on the fly.
- Requires fine-tuning to update its worldview.

This is **not** a story of **VDBs vs LLMs**



This **is** a story of **VDBs + LLMs**



This **is** a story of **VDBs + LLMs**

Focus on content:

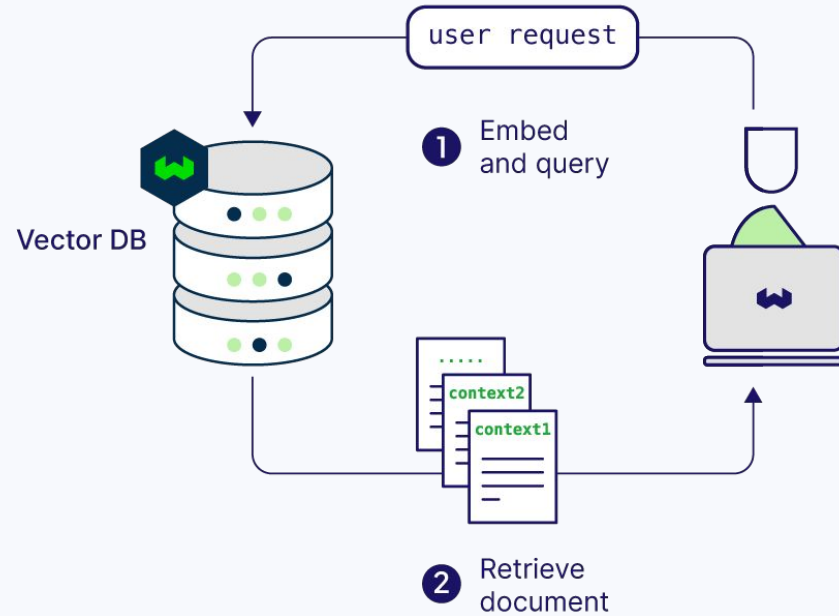
- Keep the data up to date
- Provide accurate context to LLMs



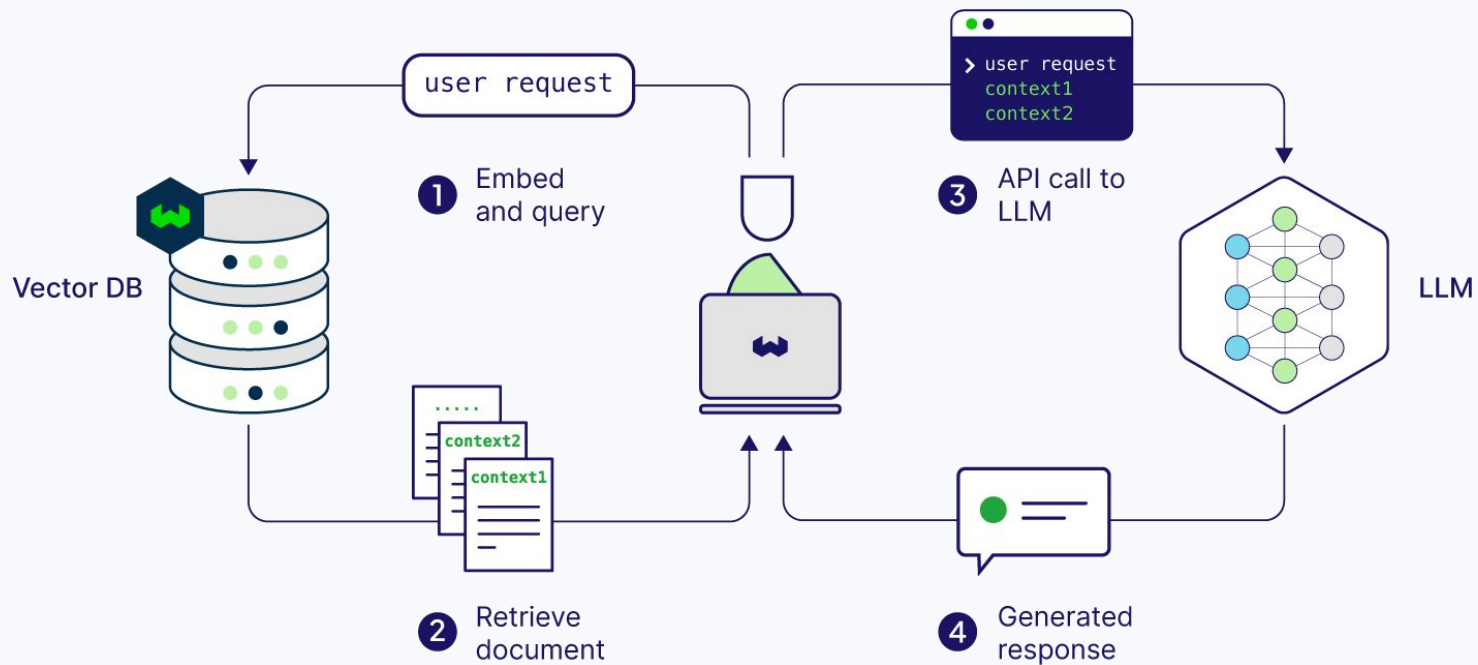
Focus on response:

- Speak with your brand voice
- Boost understanding of the underlying context

Retrieval (R)



Retrieval-Augmented Generation (RAG)



RAG – grouped task



```
1 reviews = client.collections.get("WineReview")
2 response = reviews.generate.near_text(
3     query="a sweet German white wine",
4     limit=2,
5     target_vector="description",
6     grouped_task="Summarize these reviews",
7     generative_provider=GenerativeConfig.openai(),
8 )
9
```

RAG – single prompt



```
1 reviews = client.collections.get("WineReview")
2 response = reviews.generate.near_text(
3     query="a sweet German white wine",
4     limit=2,
5     target_vector="description",
6     single_prompt="Translate this into German: {review_body}",
7     generative_provider=GenerativeConfig.openai(),
8 )
9
```


RAG – provider options

```
1 response = collection.generate.near_text(  
2     query="A holiday film",  
3     limit=2,  
4     grouped_task="Write a tweet promoting these two movies",  
5     # highlight-start  
6     generative_provider=GenerativeConfig.anthropic(  
7         # # These parameters are optional  
8         # base_url="https://api.anthropic.com",  
9         # model="claude-3-opus-20240229",  
10        # max_tokens=512,  
11        # temperature=0.7,  
12        # stop_sequences=["\n\n"],  
13        # top_p=0.9,  
14        # top_k=5,  
15    ),  
16    # highlight-end  
17 )
```

(Multi-) Tenancy

Single vs multi-tenancy architectures. Use cases for each, and how to manage resources



Single tenant

When your use case requires all data in a single collection

- Store and search through **Multi-billion scale** collections
- Process millions of updates on a daily basis
- Smart (re) vectorization techniques
- Combine keyword, and vector search for powerful **hybrid search**
- Sharding and replication allows for tuning throughput
- Fast filtering capabilities to narrow down results



Multi tenant

When your use case needs to isolate data per user

- Scales to **millions** of tenants
- Full isolation
- Tenants can be **Cold, Warm, or Hot**, depending on activity
- Control tenant status through APIs or automatically
- Store Frozen tenants on cloud storage to reduce cost
- Allows for on-disk tenants and in-memory depending on use case



Split Data per (Domain)

User

Each user with their own data

Customer

White labelled-solutions

Team / Department

Separate HR docs from legal docs



Split Data per (Resource)

PDF

Tenant for PDF chunks

Video

Tenant for Video chunks

Group of Documents

Tenant per topic

Videos

Tenant per set of videos
(i.e. course lessons)



Split Data per (Time)

Daily

Tenant per day

Monthly

Tenant per month



Built-in Multi-tenancy

Smaller Indexes

Index per tenant

Faster queries

Less to search

GDPR compliant

For User tenants

Tenant memory management

Hot and Cold

Flat Index – friendly

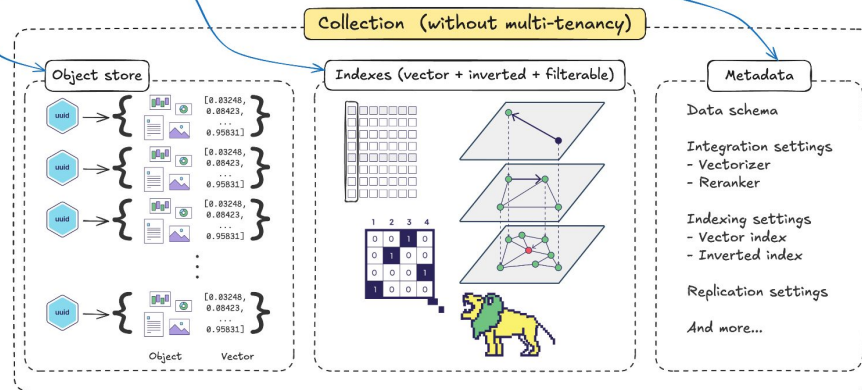
Save RAM



Each collection stores
source objects + vectors

Each collection has
sets of indexes

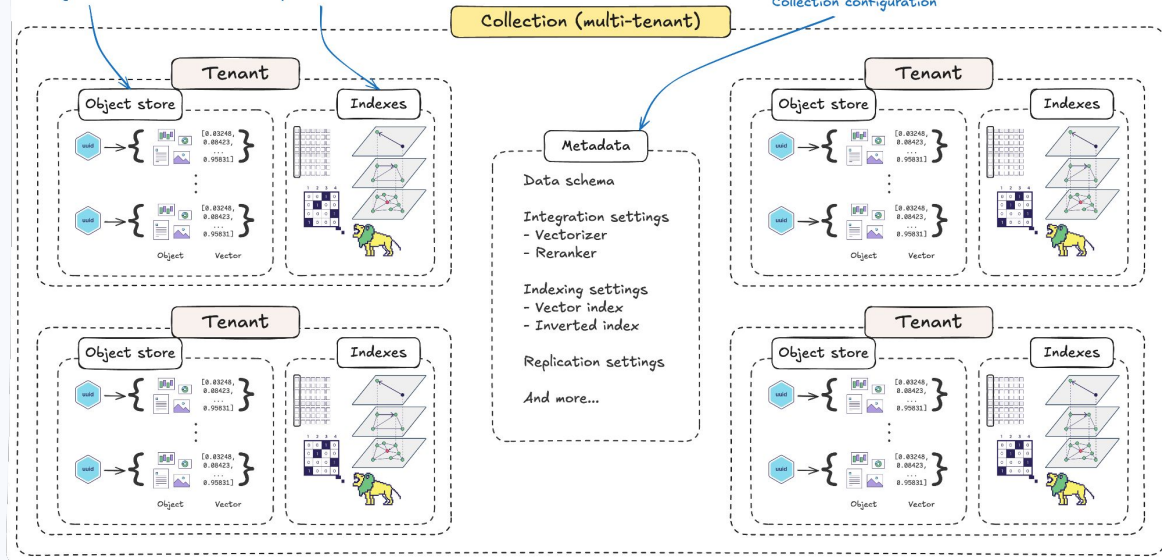
Each collection has its own
metadata / configuration



Each tenant has its own
object & vector store

Each tenant's indexes are
separate from the others

Tenants share the metadata /
Collection configuration



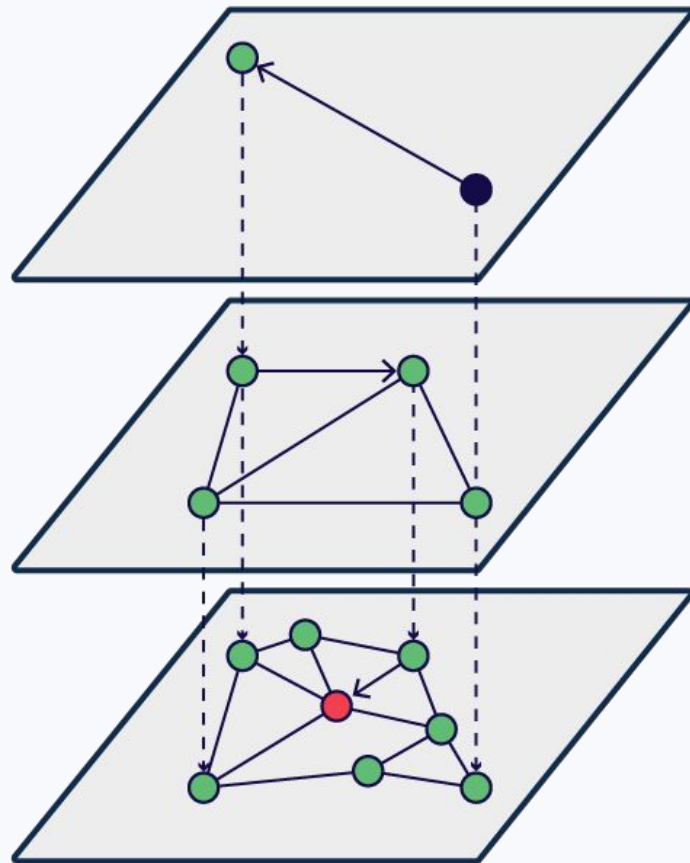


Quick notes

Weaviate at a high level +
Features we didn't cover

Indexing techniques

- **HNSW:** high-performance, in-memory. Scales well, even for very large data sets.
- **Flat:** disk-based indexes that perform brute-force vector searches. Best for small data sets.
- **Dynamic:** Automatically switches from flat to HNSW when a collection (or tenant) reaches a threshold size.





Quantization techniques

- **Product quantization (PQ):**
customizable, centroid-based quantization technique to reduce number of dimensions & precision
- **Scalar quantization (SQ):**
Converts each dimension into a scalar (integer)
- **Binary quantization (BQ):**
Converts each dimension into a bit (binary)

Product Quantization
(Available for HNSW indexes only)

[0.83, -0.27, 0.64, ..., -0.72, 0.15, -0.58]



Each dimension
becomes an integer,
and there are
fewer dimensions

[20, ..., 17]

Scalar Quantization
(Available for HNSW indexes only)

[0.83, -0.27, 0.64, ..., -0.72, 0.15, -0.58]



Each dimension
becomes an integer

[232, 95, 217, ..., 36, 146, 53]



🔍 1024 gib



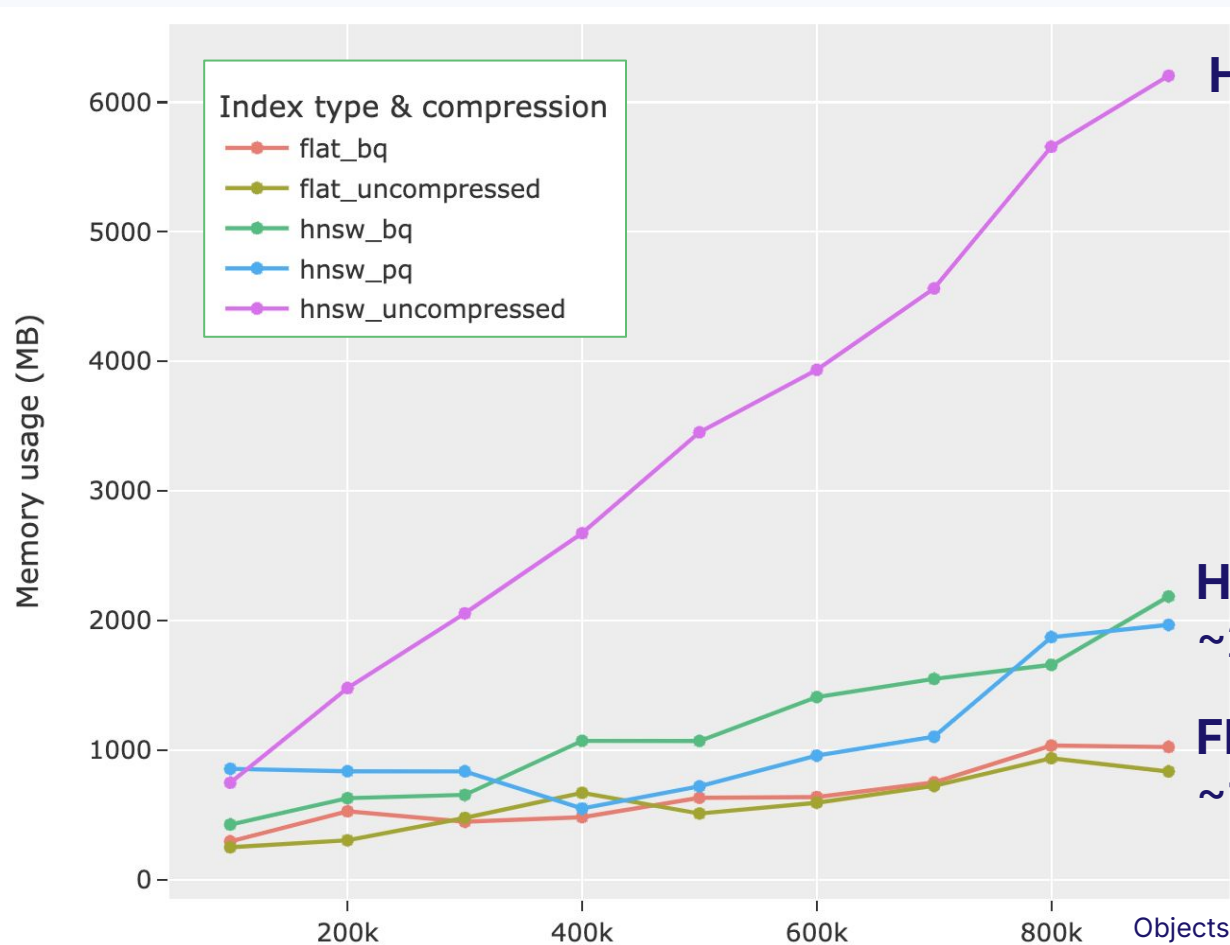
19 matches

Instance name ▾	On-Demand hourly rate ▲	vCPU ▾	Memory ▾	Storage ▾
x2gd.metal	\$5.344	64	1024 GiB	2 x 1900 SSD
x2gd.16xlarge	\$5.344	64	1024 GiB	2 x 1900 SSD
hpc6id.32xlarge	\$5.70	64	1024 GiB	4 x 3800 NVMe SSD
x2iedn.8xlarge	\$6.669	32	1024 GiB	1 x 950 NVMe SSD
x2idn.16xlarge	\$6.669	64	1024 GiB	1 x 1900 NVMe SSD

AWS Spot pricing, Nov 2024



Example: Index type & quantization vs memory footprint



HNSW: ~6GB

HNSW + BQ/PQ:
~2GB

Flat (+BQ):
~1GB



Instance type

vCPU

32

Memory

\$58k / year**vs.****\$23k / year**

View



< 1 2 >

Instance name

Demand
hourly rate

vCPU

Memory

Storage

Network
performance

x2iedn.8xlarge

\$6.669

32

1024 GiB

1 x 950 NVMe
SSD

25 Gigabit

x1e.8xlarge

\$6.672

32

976 GiB

1 x 960 SSD

Up to 10 Gigabit

x2gd.8xlarge

\$2.672

32

512 GiB

1 x 1900 SSD

12 Gigabit

AWS Spot pricing, Nov 2024

Multi-Vector Embeddings



What?

Multi-vector embeddings allow objects to be represented by multiple vectors instead of a single vector.

This approach, like Colbert, preserves more nuanced meaning by representing each token or word with its own vector, rather than compressing everything into one vector.

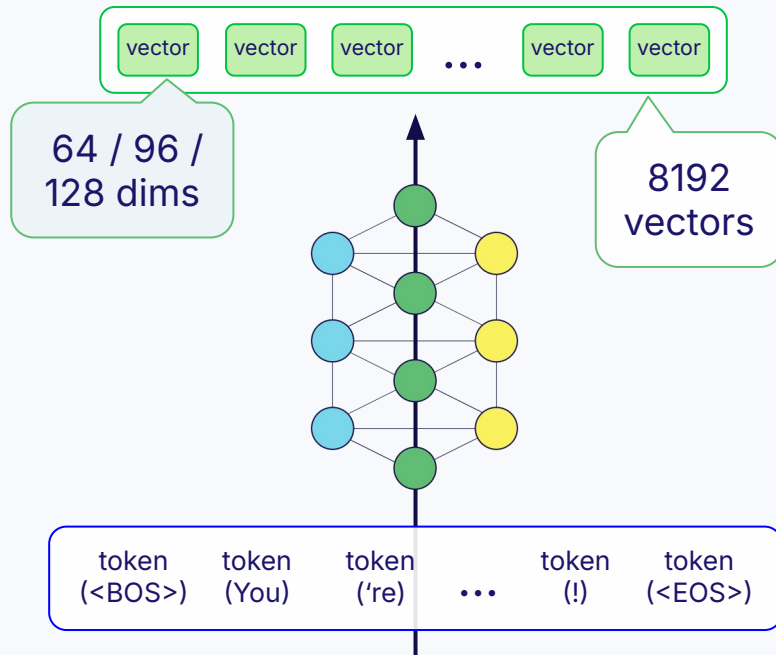
They generally provide better results but have higher memory and computational overhead.

Why?

Multi-vector embeddings provide more accurate and context-aware search results, especially for densely packed or nuanced text.

They capture finer details and relationships between words, leading to improved performance in tasks like semantic search, particularly for complex queries or specialized domains like medical data or research papers.

Multi-vector



You're a wizard, Harry!

What?

BlockMax WAND is a performance optimization technique for keyword search that significantly improves search speed.

How it works?

The algorithm organizes parts of the keyword index with pre-computed statistics, allowing it to skip multiple documents that won't be in the top results.

Why?

Weaviate implemented BlockMax WAND to significantly improve keyword search performance by reducing latency and storage requirements by efficient document skipping and data compression techniques.

BlockMax WAND

**How Weaviate
Achieved 10×
Faster Keyword
Search**



Fully Supported Role-Based Access Control



What?

RBAC is a method of **regulating access to data** in databases and collections based on the roles of individual users within an organization. It provides an additional layer of data security beyond the native multi-tenancy separation of concerns in Weaviate.

Why?

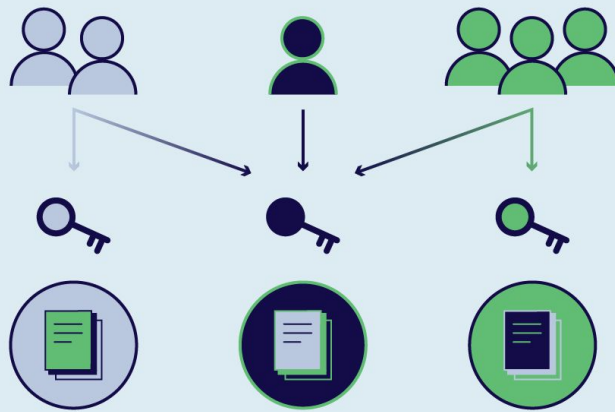
Weaviate previously offered only *readonly* or *admin* roles by default. To enable customers to create more custom roles with fine-grained permissions, RBAC is essential. This is particularly **important in enterprise ecosystems, where security, compliance, and separation of concerns are critical**.

Release Timeline

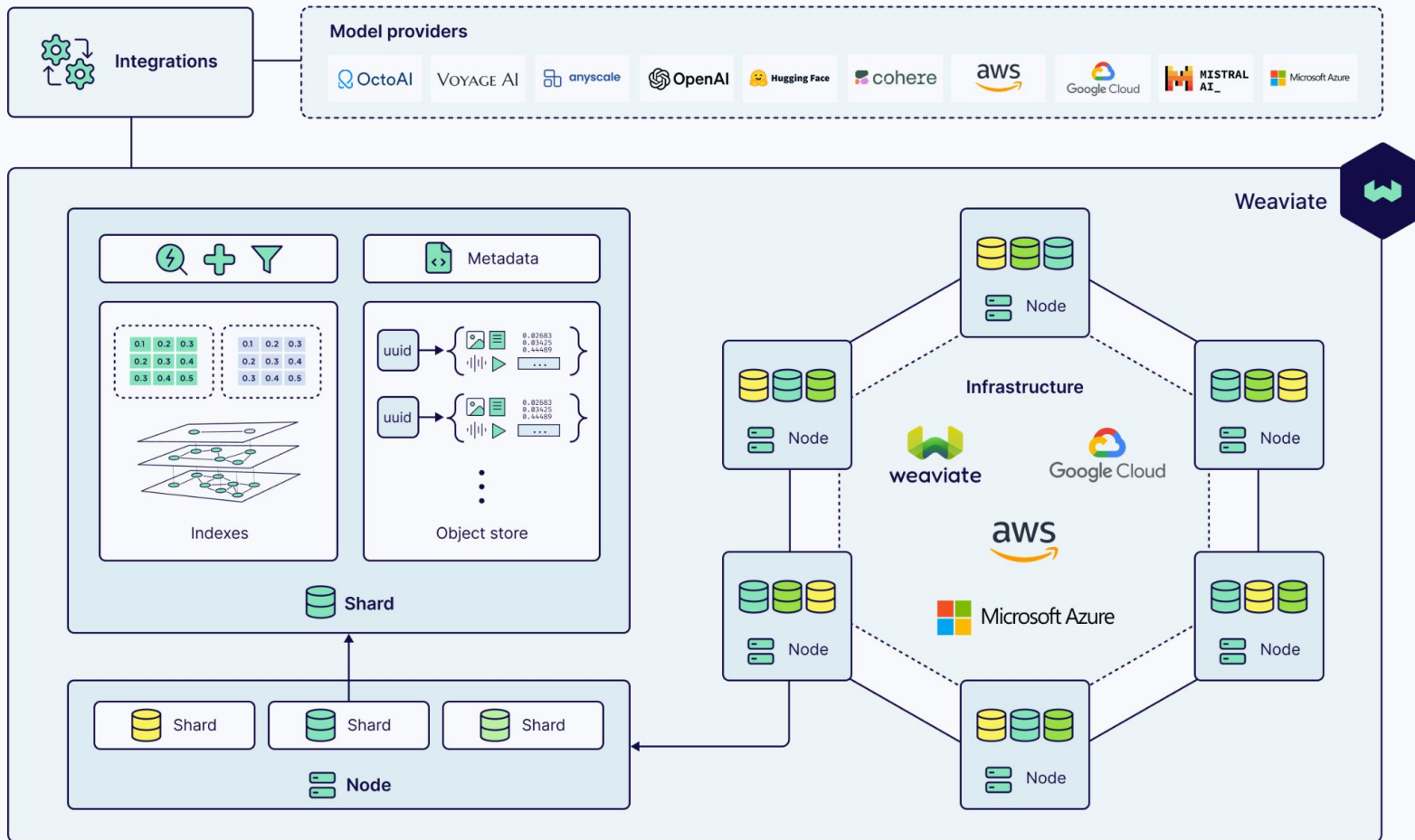
Weaviate 1.27 – Design Finalized (Q3 2024)

Weaviate 1.28 – Preview (Dec 11, 2024)

Weaviate 1.29 – General Availability (Feb 25, 2025)



Weaviate Architecture





Q&A

Ask us anything!



Thank you



weaviate.io



[weaviate/weaviate](https://github.com/weaviate/weaviate)



[linkedin](https://www.linkedin.com/company/weaviate)



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